

# Model Paper - Database Management Systems

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## Q1 (20 marks)

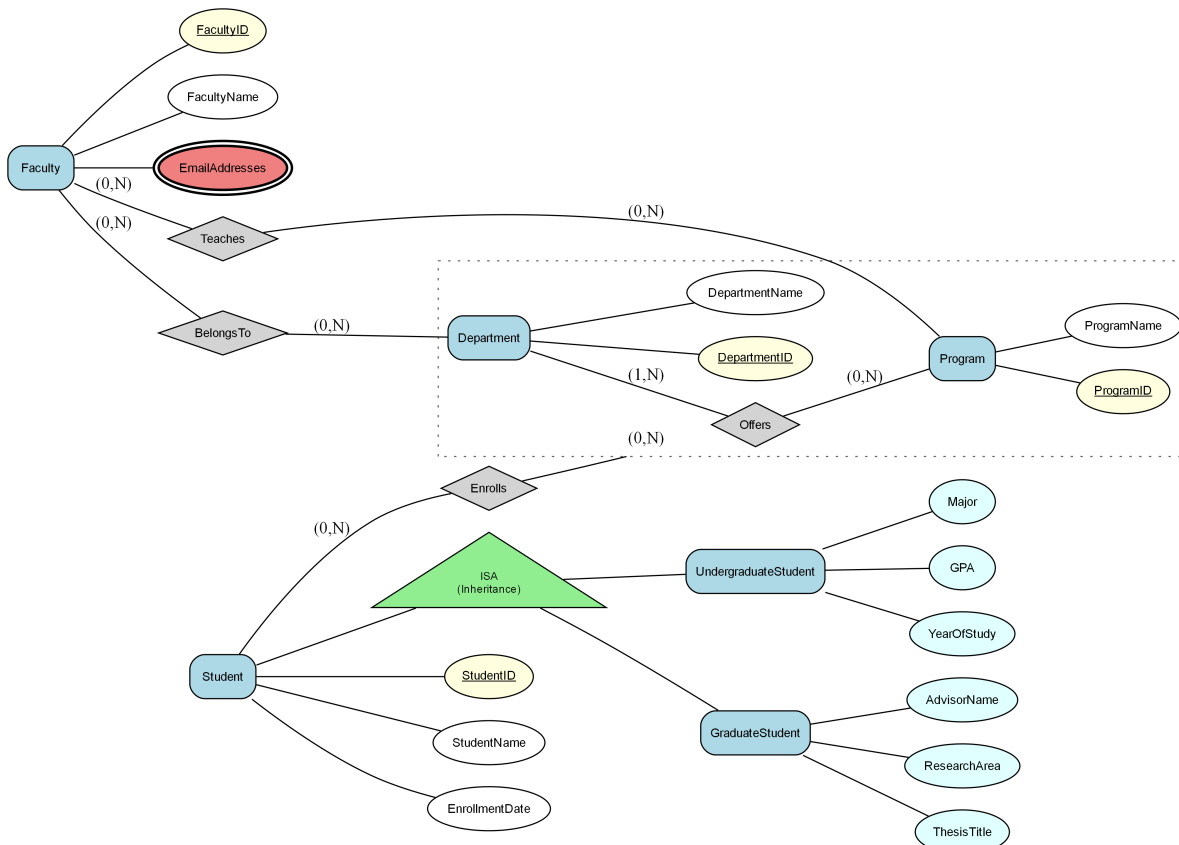
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,\*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,\*) indicates optional many participation (zero or many)

A university maintains a database to manage its academic programs, departments, and students. The university has several departments including Computer Science, Mathematics, and Physics. Each department offers multiple courses, and each course has attributes such as CourseID, CourseName, and Credits. Students can enroll in courses, and each enrollment captures relevant details such as EnrollmentDate and Grade. The university also has an ISA relationship where The Student entity has attributes such as StudentID, Name, Email. has two subtypes: UndergraduateStudent, with attributes like YearOfStudy and Major, and GraduateStudent, with attributes like ThesisTitle and Advisor. Each student can have multiple contact numbers, forming a multivalued attribute

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

## Q2 (15 marks)

Consider a relation  $R(A, B, C, D, E)$  with the following set of functional dependencies  $F$  over  $R$ :  
 $F = \{AB \rightarrow C, B \rightarrow D, CD \rightarrow E, A \rightarrow B, D \rightarrow E\}$

- a) Find all the keys in relation  $R$  using the attribute closure method. (4 marks)
- b) Is  $R$  in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is  $R$  in BCNF? Give reasons for your conclusion. If  $R$  is not in BCNF, convert it to a set of BCNF relations. (8 marks)

## Q3 (25 marks)

A university is developing a database to manage its course enrollments, student records, and faculty details. The courses have a unique identifier, title, and credits. Each student has a unique ID, name, age, and major, while faculty members have an ID, name, and department. The database should also enforce that each student can enroll in multiple courses and each course can have many students enrolled. Integrity constraints need to be specified to ensure the database remains consistent.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

- b) Code Segment:

```
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

- c) Briefly explain the options available for transferring data between tables and servers. (2 marks)
- d) Briefly explain how integrity constraints can be implemented in SQL Server. (3 marks)
- e) **A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.**

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases. (3 marks)

#### **Q4 (40 marks)**

Consider the following schema of a database designed for a university: Student (studentId: int, name: varchar(100), email: varchar(50), enrollmentDate: date) Course (courseId: int, title: varchar(100), credits: int, department: varchar(50)) Enrollment (enrollmentId: int, studentId: int, courseId: int, enrollmentDate: date, status: varchar(20)) The 'Student' table stores information about students enrolled at the university, including unique student ID, name, email, and enrollment date. The 'Course' table contains information on courses offered by the university, including unique course ID, title, credits awarded, and the associated department. The 'Enrollment' table tracks the enrollment of students in courses, including unique enrollment ID, student ID, course ID, enrollment date, and status of the enrollment. The 'Enrollment' table manages student course enrollments, recording which students are enrolled in which courses and when. Fee (feeId: int, studentId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

##### **a) Write SQL Queries to perform the following:**

- i. Find the name and email of students who are enrolled in a course titled 'Database Systems'. (4 marks)
  - ii. Find the student with the highest aggregate value using GROUP BY on related enrollment records. (6 marks)
  - iii. Find the title and name from Course, Student, and Enrollment where Enrollment.enrollmentId is not null. (7 marks)
- b) Create a function to calculate the total fee amount for a given student. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)
- c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each student. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by student. The student pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)